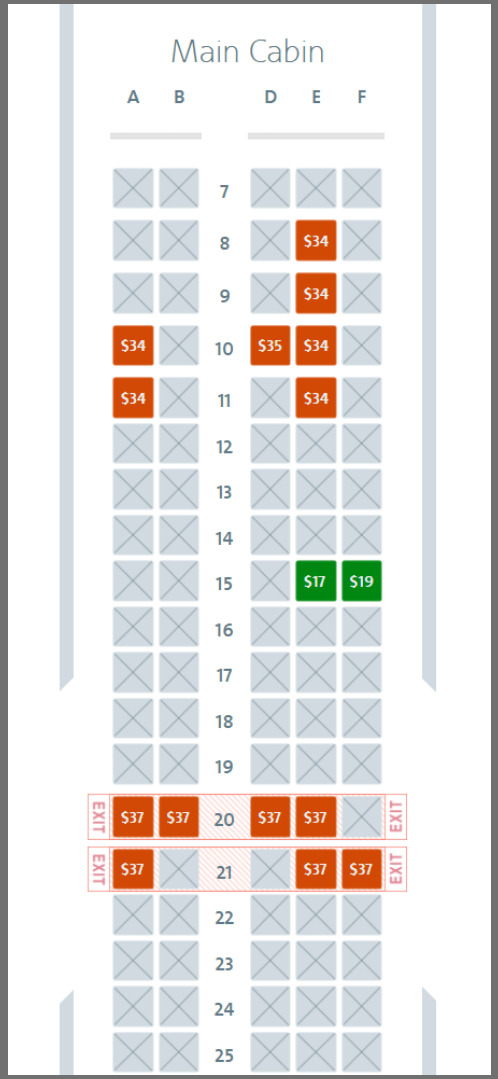


Transactions Overview



Motivation

Joe:

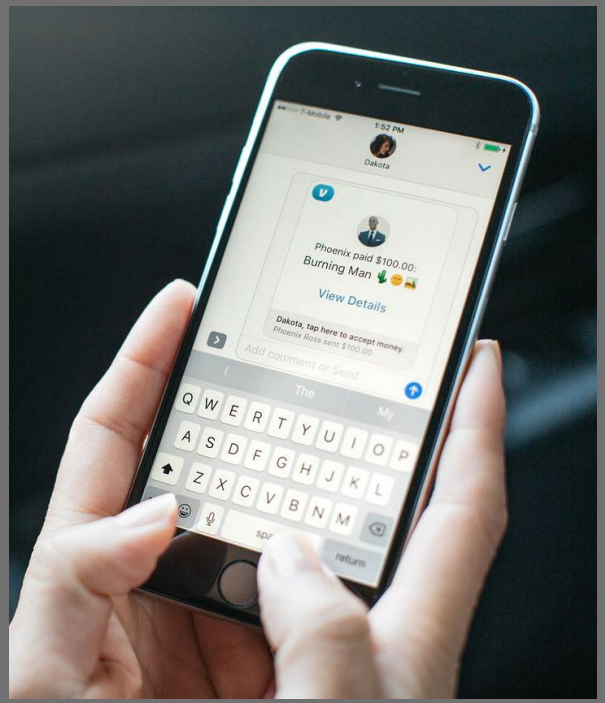
- Show available seats
- Choose 15F
- Confirm available 15F
- Buy 15F

Sue:

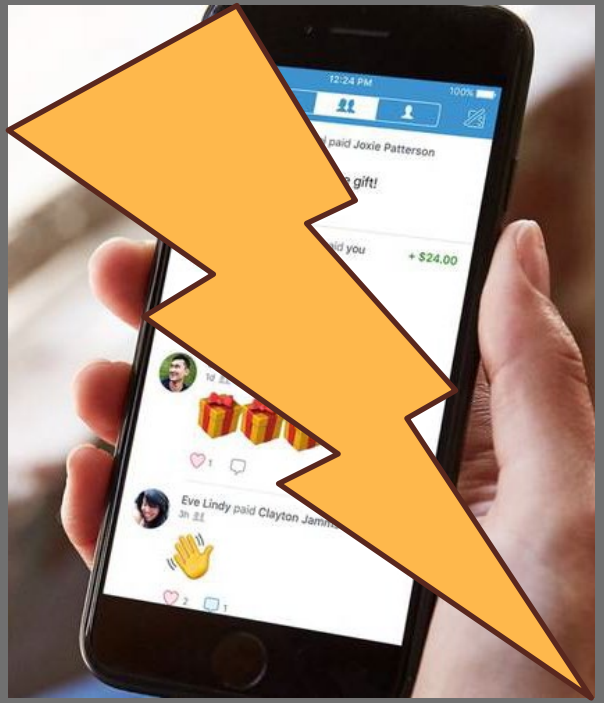
- Show available seats
- Choose 15F
- Confirm available 15F
- Buy 15F

Seat double-booked!

Motivation







Need to rollback transaction.

Motivation



ACID properties

Atomicity	• All-or-nothing execution
Consistency	• Always in valid state
Isolation	• Concurrent users without conflict
Durability	• Permanent data

Concurrency and recovering from failure are important in any modern system.

How to create a transaction

```
WITH
    …
SELECT …
FROM (SELECT …) AS Data

UNION

SELECT …
FROM …;
```

```
BEGIN TRANSACTION;
    …
    …
    …
COMMIT TRANSACTION;
```